1 FIG-217 — Same-configuration force disagreement across MLIP checkpoints

Group: MD-FROM-SCRATCH-FAILURE-ATTRIBUTION **Evidence chain:** manifested **Status:** paired force-disagreement diagnostic; calibration pending

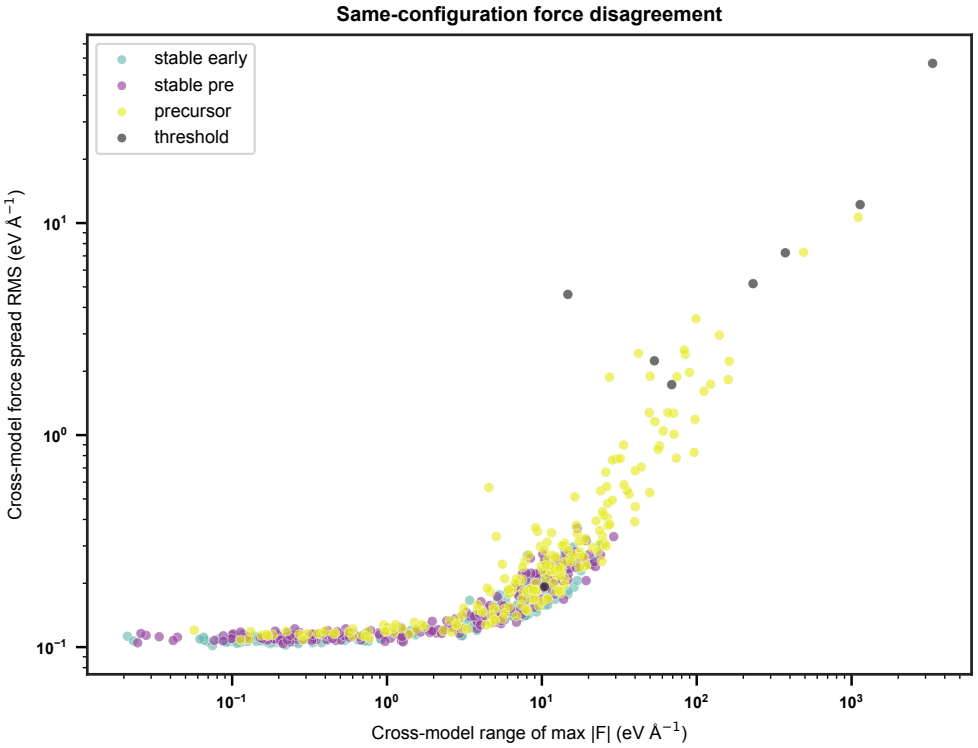


Figure 1: Cross-model force spread versus the cross-model range of maximum force for identical candidate structures. Points are colored by source stage. Both axes are logarithmic to retain the stable controls and the threshold tail; the relationship is a descriptive paired-model diagnostic, not a calibrated uncertainty estimate.

Results text draft. The median force-spread RMS is 0.121 eV Å^{-1} for stable-early controls, 0.131 eV Å^{-1} for stable-pre candidates, 0.229 eV Å^{-1} for precursors, and 5.17 eV Å^{-1} for threshold-or-terminal candidates. The corresponding 95th percentiles are 0.269, 0.265, 1.88, and 43.1 eV Å^{-1}, respectively. **Discussion and boundary.** The paired rise in disagreement toward precursor and threshold structures is compatible with model-dependent amplification near the instability boundary. Because the checkpoints share training lineage and no independent force reference is available yet, the spread cannot be interpreted as a calibrated error bar, a proof of extrapolation causality, or evidence that a specific model is correct. **Article placement.** Proposed placement: Supplementary Information / paired MLIP force-disagreement diagnostic; it should not be described as uncertainty calibration. **Reviewer linkage:** overall status PENDING.

- **REV-C9 (PENDING)**: Assess the use of paired force disagreement to prioritize independent calculations. *Completed:* same-configuration comparisons and stage-resolved quantiles. *Still required:* reference-force calibration and model-lineage sensitivity analysis.

Evidence pointers.

- Paired spread table: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/results/candidate_model_spread.csv
- Pairwise force table: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/results/pairwise_force_disagreement.csv
- Plot script and Plot Atlas source: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/make_mlip_analysis.py; /Users/alina/Documents/Codex/plot_library

- Asset SHA256: `f96a685965581d5ce1c0b26debf007f735530950908342bd95bcfad70c0ae1d4`

Machine-collected evidence candidates (generated from the analysis manifest; do not edit this file).

- **Analysis manifest:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/results/analysis_manifest.json
- **Plot output manifest:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/figures/figure_and_text_manifest.json

Audit gap / next action: Use the selected 120-atom PBC reconstructions for converged QE force comparisons; preserve the distinction between model disagreement and DFT error.